VANSH LAKHWANI

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Profile Summary

Aspiring AI/ML Engineer with a strong foundation in software development, data preprocessing, and model deployment. Seeking a role in the artificial intelligence and machine learning domain, with hands-on experience in using Python, Scikit-learn, TensorFlow, NumPy, and Pandas for developing supervised and unsupervised learning models.

Education

- **Bachelor of Technology –** Artificial Intelligence and Data Science, CGPA: **9.42** **Expected - 2026**
  Dr. D.Y. Patil School of Science and Technology, Pune, India
- **Senior Secondary Education** , Percent : **86.2%** **July - 2022**
  LifeLine Public School, Agra, India

Experience

Uluka Systems Pvt. Ltd. , Machine Learning Intern **August 2024 - Feb 2025**

- Built an **ML-powered Road Condition Assessment** model using real-time sensor data (accelerometer, gyroscope, GPS) collected via the **Sensor Logger app** on Android smartphones.
- Executed **K-Means clustering** to classify road surfaces into **smooth, rough, and mixed conditions**, using statistical features like total acceleration and speed.
- Performed **data preprocessing** and **feature engineering** on multi-sensor time-series data, merging 13 CSV datasets, handling missing values, and applying PCA for dimensionality reduction.
- Visualized clustered road conditions on an interactive **Folium map**, enabling intuitive geographic insights for infrastructure planning and hotspot detection.

Projects

- **Smart CCTV Surveillance System -** *Python, OpenCV, Machine Learning*
  Developed an ML-powered surveillance system leveraging existing CCTV infrastructure for crowd management, criminal detection, face recognition, and real-time alert messaging to enhance public safety and reduce human oversight.
- **Instacart Market Basket Analysis in R -** *R, Data Visualization, Association Rule Mining*
  Examined Instacart transaction data to uncover product association patterns and recommend strategies to boost customer retention and product bundling for increased sales.
- **Instacart Market Basket Platform (Django) -** *Python, Django, HTML/CSS*
  Built a full-stack e-commerce simulation platform using Django to provide a seamless grocery shopping experience. Executed user login, product browsing, and predictive recommendations.
- **Movie Ticket Booking System -** *C Programming*
  Designed a console-based, user-friendly system to simulate real-time movie ticket booking, including seat selection, ticket availability, and booking confirmation.
- **Little Mumkins E-commerce Website (Ongoing) -** *WordPress, WooCommerce, Elementor*
  Currently designing and optimizing an e-commerce platform for 'Little Mumkins', a clothing brand. Responsibilities include UI customization, performance tuning, and streamlining the online purchase process for better user experience.

Certificate

- **Web Development Certification**
- **AWS online Conference**
- **Machine Learning in Surveillance**
- **Django Workshop**

Key Skill

Technical Skills :

- ***Programming Languages*** : Python, C, C++, R
- ***ML/DL Libraries*** : Scikit-learn, tensorflow
- ***Data Handling/Analysis*** : Pandas, NumPy, Matplotlib
- ***Tools/Platforms*** : Git, Jupyter, VS Code, Google Colab, Kaggle
- ***Databases*** : SQL

Soft Skills : Project Management | Collaboration skills | Leadership skills | Resilient | Continuously Upskilling | Analytical Mindset

Award and Achievement

- **Team Lead, Smart India Hackathon**
- **3rd Place, 5G Nexus Ideathon**
- **Club Lead, Binary Brain Coding Club**